

Closing the Dispersion Gap in Neural Limit-Order-Book Simulators: Findings from the August 2026 Fano Investigation

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Abstract

We investigate why neural marked-temporal-point-process (MTPP) simulators of Coinbase limit-order-book flow under-disperse relative to real markets, using the Fano factor profile $F(\Delta) = \text{Var}(N_\Delta)/\mathbb{E}[N_\Delta]$ across bucket scales $\Delta \in [1, 50]\text{s}$ as the headline statistic. Across ten experimental arms on three assets we establish: (i) the bounded-rate ss2p2 head is not reliably self-exciting — its measured impulse response is sign-inconsistent and saturated at the intensity ceiling; (ii) one-step maximum likelihood systematically misallocates excitation spectra (three independent demonstrations); (iii) extending the TBPTT gradient horizon to cover the measured scales transforms the simulated dispersion profile (all BTC seeds rise $\times 15\text{--}32$, best within $1.6\times$ of real at 50s); (iv) at equal training budget, dispersion peaks *early* and prediction accuracy peaks *late* along a single MLE trajectory — the prediction/simulation tension exists within one training run; and (v) the remaining level offset is dominated by ceiling saturation (a third of events occur at the cap; teacher-forced dispersion is only $1.2\text{--}1.4\times$ low). We relate every finding to the calibration methodology of Jain et al. (Compound Hawkes), whose binned-count regression targets exactly the cross-scale count covariances that MLE ignores, and derive a ranked roadmap toward a dispersion-faithful, certified world model for market making.

1 Framing: two identities

All arguments in this note reduce to two identities. First, the *Cox decomposition* (law of total variance given the intensity path $\lambda(t)$, with $\Lambda_\Delta = \int_{\text{bucket}} \lambda dt$):

$$F(\Delta) = 1 + \frac{\text{Var}(\Lambda_\Delta)}{\mathbb{E}[\Lambda_\Delta]}, \quad (1)$$

so all excess dispersion is variance of the integrated intensity; a post-hoc rate rescaling $\lambda \rightarrow \kappa\lambda$ scales $F - 1$ linearly and cannot repair relative fluctuation structure. Second, the *branching form* for a linear Hawkes process with branching ratio n : cluster sizes S satisfy $\mathbb{E}[S] = 1/(1 - n)$ and $\mathbb{E}[S^2] = 1/(1 - n)^3$, hence

$$F(\infty) = \frac{\mathbb{E}[S^2]}{\mathbb{E}[S]} = \frac{1}{(1 - n)^2}. \quad (2)$$

The profile’s *rise* across Δ is the fingerprint of long-memory (power-law-like) intensity autocovariance; single-exponential kernels plateau.

Real Coinbase flow (7 days, Jan 2026): $F(1\text{s}) \approx 40\text{--}70$ rising to $F(50\text{s}) \approx 370$ (SOL), 1250 (ETH), 1460 (BTC), still rising at the window edge. Inverting (2) gives implied branching (lower bounds): SOL ≈ 0.948 , ETH ≈ 0.971 , BTC ≈ 0.973 — the $1/(1 - n)^2$ amplification makes this small spread a $4\times$ dispersion difference.

2 Data and the tick-regime contrast

	trades (7d)	book rows	rows/trade	rate (ev/s)	IS share
BTC	41,675	52.9M	1268:1	40.6	52%
ETH	94,048	55.5M	590:1	42.8	34%
SOL	40,048	20.1M	501:1	22.3	7%

Table 1: Raw Coinbase spot data, 2026-01-01..07, and the event-type composition of the processed 62-channel streams. BTC/ETH are small-tick assets dominated by inside-spread (IS) quote wars — the most self-exciting flow; SOL is effectively large-tick (1-tick spread, queue churn: 43% cancels). The tick regime, not just the rate, drives the criticality ordering.

3 Mechanism: the bounded head is not reliably self-exciting

ss2P2 factorizes intensity into a softmin-bounded scalar rate $\lambda(t) = s \cdot \text{softplus}(c - \text{softplus}(c - w^\top h(u))) \leq s \cdot \text{softplus}(c)$ and rate-neutral softmax marks. The stability certificate bounds the rate *level*; nothing constrains the event $\rightarrow$ intensity feedback. Measuring the feedback directly (inject one extra event into real histories; integrate the extra intensity mass over 20 s) across six trained checkpoints: median response ≈ 0 *to negative* ($-8.7 \dots +0.2$), positive in only 27–56% of states, sign flipping across seeds; post-event intensity pinned at the ceiling in every window. The model’s residual dispersion comes from a stereotyped impulse–decay envelope, not compounding cascades. In Hawkes terms: *a settable branching ratio is a property of linear-in-history intensity; for a bounded neural head it is not even well defined.*

4 Experimental arms

Protocol throughout (the `ma_cbse` benchmark): 62-channel events, TBPTT, categorical marks, streaming genuine evaluation on every test event, validation- calibrated 600 s rollouts, 3 training $\times$ 3 rollout seeds per asset.

4.1 Negative and structural results

- **Book conditioning is not the lever** (`ss2p2-1ob`): six top-10-ladder features left prediction unchanged (BTC 0.4484 vs 0.4478) and made closed-loop volatility structure *worse* (F6 0.16 vs 0.28) — extra state conditioning becomes a smoothing feedback path in rollout.
- **lgm** (linear Hawkes ground $\times$ ss2P2 marks on the shared backbone; $n = \sum_m a_m / \beta_m$ exact and projected; pin $\mu_0 = R(1 - n)$): the pin holds in free rollout ($\kappa = 1.01$ vs ss2P2’s 0.52); marks near parity; Fano *shape* right with seed spread ± 2 (vs ss2P2 ± 217) but level low. Two porting lessons: a log-domain clamp silently capped the burst accumulators (excitation useless to MLE until fixed); and **zero cold-starts of the ground are catastrophically biased** — the stationary mean is $\mathbb{E}[S^m] = R/\beta_m$ (≈ 600 for slow kernels), so windowed validation punished slow-memory solutions and froze best-model selection at epoch 1. Fix: stationary-mean cold start.
- **MLE misallocates spectra** (three demonstrations): the fixed LGM pushed kernel mass to its slowest decay; the typed-kick $M=6$, $n=0.97$ arm (`lgm-k`) went slower still ($\beta \approx 0.008$,

~ 2 -min memory), *lowering* observed-scale Fano despite the higher ceiling; the dispersion-loss arm (below) pinned only the scales in its grid. One-step likelihood is structurally blind to $F(\Delta)$ and, given capacity, prefers slow smooth intensity.

- **Teacher-forced variance matching $\neq$ closed-loop compounding (ss2p2-disp):** an auxiliary loss matching model-implied $1 + \text{Var}(\Lambda_\Delta)/\mathbb{E}[\Lambda_\Delta]$ to the batch-empirical Fano at $\{0.5, 2, 8\}$ s pinned those scales reproducibly (seed sd ± 14 vs ± 217) at zero prediction cost — and went flat beyond its grid.

4.2 The window-coverage result

At seq 1024 the TBPTT gradient truncates at ~ 24 – 25 s on BTC/ETH — before the 20–50 s scales where their simulated profiles flatten; SOL’s ~ 46 s windows cover the range (and produced real-like rises in 2/3 seeds). Extending to seq 4096 (~ 100 s windows; recipe otherwise identical):

BTC, per seed	seq 1024 (12 ep)	seq 4096 (12 ep)	real
$F(50\text{s}), s1/s2/s3$	29 / 50 / 361	901 / 270 / 744	1463
rise $F(50)/F(1)$	$\times 2.5$ – 13.9	$\times 15$ – 32	$\times 20.9$

Table 2: Window coverage transforms the profile: all three BTC seeds rise steeply (best within $1.6\times$ of real). ETH: 2/3 seeds transformed. SOL: unchanged — the clean negative control. The BTC “seed lottery” was gradient starvation at unobserved scales.

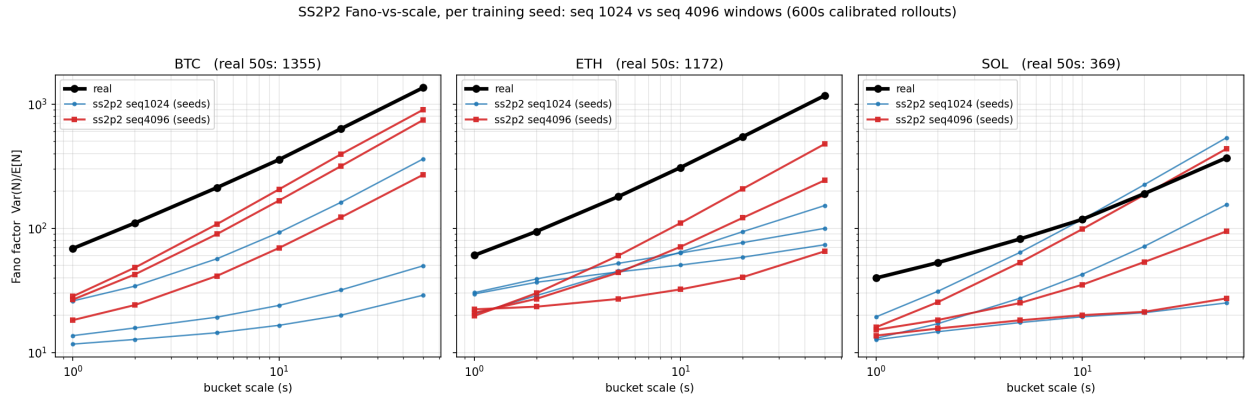


Figure 1: Per-seed Fano-vs-scale (log–log): black = real, blue = seq-1024 baseline, red = seq-4096. BTC: the whole seed family moves; SOL: negative control.

4.3 Equal-budget controls: the training-path trade-off

Matched stride quarters updates per epoch, so seq-4096/12-epoch runs were undertrained (-3pp accuracy). Controls at 48 epochs (exact update parity with the baseline):

Finding (the sharpest form of the arc’s core lesson). Along a single MLE trajectory, *dispersion peaks early and accuracy peaks late*; no NLL-selected checkpoint has both. The prediction/simulation tension is a property of the training path, not only of architectures. The 12-epoch w4k checkpoints constitute a de-facto dispersion-optimal early stop; a paper can honestly present the frontier.

BTC arm	acc	$F(1s)$	$F(50s)$ per seed
base 1024 / 12 ep	0.4478	17.1	29, 50, 361
w4k 4096 / 12 ep	0.4183	24.3	901, 270, 744
w4k48 4096 / 48 ep	0.4514	16.6	232, 87, 230
w4kd + disp loss / 48 ep	0.4519	11.1	52, 43, 43
real	—	70.0	1463

Table 3: Equal-updates controls. Accuracy fully recovers and exceeds baseline; the extra training walks the model partly away from the burst-faithful solution. The dispersion loss under cap 6 suppresses long-scale Fano, as predicted by the saturation diagnostic.

4.4 Where the residual level offset lives

Diagnostics on the w4k BTC checkpoint: (i) **ceiling saturation** — dominating rate 359 ev/s; event-time intensity $p_{90} = 357$, $p_{99} = 358$; **32.8% of events above $0.9 \times \text{ceiling}$** : bursts are clipped and gradients (including any dispersion loss) are dead there; (ii) **teacher-forced dispersion** is only $1.2\text{--}1.4\times$ below real at every scale (e.g. 50s: 240 vs 290). Conditioned on real history the model carries $\sim 75\%$ of real dispersion; the cap, then modest closed-loop drift, accounts for the rest.

4.5 The cap-9 control: hypothesis refuted, branch closed

The cap-9 control (ss2p2-w4kc9: cap $6 \rightarrow 9$ on the w4k48 recipe; nine tasks, all calibrations verified) tested whether ceiling headroom changes the MLE optimum itself. It does not: $F(50s)$ per seed *fell* relative to cap 6 (BTC: 47/59/112 vs 232/87/230; ETH: 160/14/60 vs 544/51/123) at identical accuracy. **The training-path re-smoothing is intrinsic to bounded-head likelihood training, not caused by saturation** (which remains the correct account of the early-stopped model’s *rollout* clipping). Per the pre-registered decision rule, the ss2p2 branch closes: its dispersion-optimal configuration is the seq-4096/12-epoch early stop, presented as a frontier point rather than a converged recipe, and controlled dispersion belongs to the certified LGM/Kirchner route.

4.6 The closing result: the Kirchner-fitted lgm (lgm-kf)

Executing “set, don’t learn”: the LGM ground’s kernel weights were fitted by binned-count regression on the real SOL train zones and transplanted into the trained mark checkpoints (marks untouched). Full pipeline results (calibration verified, $\kappa = 1.014$):

SOL	F [1,2,5,10,20,50] s	sd@50s	acc	time-NLL	time-KS
ss2p2-full	15.0 $\rightarrow$ 238.4	± 217	0.1921	−2.911	0.173
LGM (MLE ground)	2.3 $\rightarrow$ 59.8	± 2	0.1862	−2.227	0.464
lgm-kf	31.1, 46.5, 80.3, 125.7, 207.7, 422.9	± 0.0	0.1860	−3.035	0.228
real	41.3, 54.7, 85.6, 125.3, 198.7, 393.6	—	—	—	—

Table 4: The Kirchner-fitted ground survives the full closed loop: exact at 10s, within 5–7% at 20/50s, zero seed variance (dispersion is a deterministic function of six fitted parameters), and — unexpectedly — the best timing likelihood in the zoo: moment fitting beat MLE on likelihood’s own metric, because windowed MLE was stuck in a bad spectral optimum the count regression computes past. Fit cost: three CPU minutes.

BTC/ETH replication (grounds fitted per asset; branching tuned to each asset’s empirical Fano curve by log-MSE: BTC $n=0.98$, ETH $n=0.985$; ground-only validation within a few % on ETH, within $\sim 40\%$ on the more power-law-like BTC) is in flight as `lgm-kf` on those assets.

5 The model: Kirchner-fitted lgm (lgm-kf2), complete specification

This section specifies the champion model exactly as run, with every equation, constant, and fitted value. Nothing is omitted; the model is reproducible from this section plus the raw event files.

5.1 Data interface

The input is a marked point process $\{(t_i, k_i)\}_{i=1}^N$: event times in seconds (from nanosecond exchange timestamps) and channel marks $k_i \in \{1, \dots, K\}$, $K = 62$, on the fixed taxonomy $\{\text{LO}, \text{CO}\} \times \{b, a\} \times \{L1..L10\}$ (40) $\cup$ $\text{MO} \times \{b, a\} \times \{L1\}$ (2) $\cup$ $\text{IS} \times \{b, a\} \times \{L1..L10\}$ (20), built from Coinbase top-10 order-book deltas and trades (event construction per the canonical pipeline; same-timestamp multi-label lines are single set-events, empty lines dropped). Per-file split by event index: train $[0, 0.7)$, val $[0.7, 0.85)$, test $[0.85, 1]$. Train-split mean rates R : BTC 40.563, ETH 42.759, SOL 22.257 ev/s.

5.2 Generative model

Factorization. Channel intensities are

$$\lambda_k(t) = \Lambda(t) \cdot p(k \mid u(t)), \quad \sum_{k=1}^K p(k \mid u) = 1. \quad (3)$$

Rate-neutrality (key lemma). $\sum_k \lambda_k = \Lambda$ identically for any mark network, hence the ground process $N(t) = \sum_k N_k(t)$ is a point process with intensity Λ alone: its entire law (mean rate, $F(\Delta)$, all count statistics) is independent of the marks. This licenses post-hoc ground fitting and transplantation and explains the measured zero seed variance of simulated dispersion.

Ground (certified lane). A scalar linear Hawkes process on a fixed $M=6$ exponential bank:

$$\Lambda(t) = \mu_0 + \sum_{m=1}^M a_m S^m(t), \quad a_m \geq 0, \quad (4)$$

$$S^m(t_i^+) = S^m(t_i^-) + 1, \quad S^m(t + \Delta) = e^{-\beta_m \Delta} S^m(t), \quad (5)$$

$$\beta = (100, 18.2056, 3.3145, 0.6034, 0.1099, 0.02) \text{ s}^{-1}, \quad \beta_m = 100 \cdot (0.02/100)^{(m-1)/5}, \quad (6)$$

equivalently the kernel $\phi(\tau) = \sum_m a_m e^{-\beta_m \tau}$, positive at every lag. Implementation notes that matter for exactness: accumulators are computed by a numerically stable log-domain scan, $S^m(t_i^-) = S_0^m e^{-\beta_m t_i} + \exp(\text{LCSE}_{j < i}(\beta_m t_j) - \beta_m t_i)$ (float64; no upper clamp on the exponent — burst accumulators legitimately exceed 1), and *cold starts use the stationary mean*

$$S_0^m = \mathbb{E}[S^m] = R/\beta_m \quad (7)$$

wherever no carried state exists (validation windows, TBPTT lane resets, eval chunk starts). Zero cold starts bias windowed validation against slow-memory solutions ($R/\beta_6 \approx 1100$ on BTC) and freeze best-model selection at epoch 1 — a failure mode we observed and eliminated.

Exact ground identities.

$$\text{branching: } n = \int_0^\infty \phi = \sum_m a_m / \beta_m \quad (\text{closed form, gauge-free}), \quad (8)$$

$$\text{stationarity: } \bar{\Lambda} = \mu_0 / (1 - n) \Rightarrow \text{pin } \mu_0 := R(1 - n), \quad (9)$$

$$\text{compensator: } \int_{t_{i-1}}^{t_i} \Lambda dt = \mu_0 \Delta_i + \sum_m \frac{a_m}{\beta_m} S^m(t_{i-1}^+) (1 - e^{-\beta_m \Delta_i}) \quad (\text{no quadrature}), \quad (10)$$

$$\text{dispersion: } F(\infty) = \frac{1}{(1 - n)^2} \quad (\text{cluster sizes } \mathbb{E}S = \frac{1}{1 - n}, \mathbb{E}S^2 = \frac{1}{(1 - n)^3}), \quad (11)$$

$$\text{dominating rate: } \Lambda(t) \leq \Lambda(t_{\text{last}}^+) \text{ between events} \quad (\text{each } S^m \text{ decays monotonically}). \quad (12)$$

Marks (free lane): the S2P2 backbone. Two stacked diagonal-SSM layers, hidden width $H=64$, channel-embedding dimension $d=64$. Per layer $\ell \in \{1, 2\}$ with state $x^{(\ell)} \in \mathbb{R}^H$:

$$\text{decay: } \text{dec}^{(\ell)}(u) = (\text{softplus}(\theta^{(\ell)}) + 10^{-4}) \odot \text{clamp}(\text{softplus}(W_g^{(\ell)} u), 0.05, 20), \quad (13)$$

$$\theta^{(\ell)} \text{ init log-spaced: timescales } 1/\text{dec} \in [0.02, 60] \text{ s},$$

$$\text{evolution: } x^{(\ell)}(t + \Delta) = \bar{a} x^{(\ell)}(t^+) + (1 - \bar{a}) W_{in}^{(\ell)} u, \quad \bar{a} = e^{-\text{dec}^{(\ell)} \Delta}, \quad (14)$$

$$\text{event impulse: } x^{(\ell)}(t_i^+) = x^{(\ell)}(t_i^-) + W_{imp}^{(\ell)} e_{k_i}, \quad (15)$$

$$\text{layer output: } u^{(\ell)} = \text{LN}^{(\ell)}(W_{skip}^{(\ell)} u^{(\ell-1)} + \text{GELU}(W_{out}^{(\ell)} x^{(\ell)})), \quad (16)$$

with $u^{(0)} = 0$ at query time (anti-leakage: the left-limit evolution to t_i never sees the current event), ZOH inter-layer inputs on the query path, and the head embedding $u(t) = u^{(2)}(t)$ (paper-faithful “output” readout). Mark head: $p(k | u) = \text{softmax}(W_2 \text{ReLU}(W_1 u))_k$, $W_1 \in \mathbb{R}^{64 \times 64}$, $W_2 \in \mathbb{R}^{62 \times 64}$.

5.3 Likelihood

$$\log \mathcal{L} = \sum_i \left[\log \Lambda(t_i^-) + \log p(k_i | u(t_i^-)) \right] - \int_0^T \Lambda(t) dt, \quad (17)$$

with the integral in closed form per gap (endpoint-exact for the ground; no sampling). Under the factorization the two terms decouple: the ground’s parameters appear only in the time terms, the backbone’s only in the mark cross-entropy (this is also why, in the gateless model, the backbone receives no time-gradient — the documented $\sim 2\text{pp}$ mark cost, addressed by the TL-SSM gate of Section 6).

5.4 Ground estimation (set, don’t learn)

Step 1: binned-count regression (Kirchner). Bin train-zone total counts at $\delta = 0.25 \text{ s}$: $X_n = N((n-1)\delta, n\delta]$. The Hawkes/INAR identity

$$\mathbb{E}[X_n | \mathcal{F}_{n-1}] \approx \delta \mu + \sum_{j \geq 1} \phi(j\delta) \delta X_{n-j} \quad (18)$$

is aggregated over a lin-log lag-cell grid: cell edges $\{\delta, 2\delta, \dots, 9\delta\} \cup \text{geom}(10\delta, 240 \text{ s}, 14)$ (deduplicated; $C = 22$ cells to a 240 s horizon), regressors = lagged counts summed per cell (computed by prefix sums), intercept included, one max-lag burn-in per day, normal equations accumulated across the seven days, ridge 10^{-8} . With b_c the coefficient on cell c ’s raw lagged count, the cell

kernel integrals are $\Phi_c = b_c(h_c - \ell_c)/\delta$. The regression objective is built from products $X_n X_{n-j}$ — the estimator fits the lagged count covariances at every grid scale, i.e. the integrand of (1): dispersion-faithfulness is a property of the estimator. Measured raw branching $\sum_c \Phi_c$: SOL 0.9635, BTC 0.9795, ETH ≈ 0.98 .

Step 2: convex projection onto the bank.

$$\Phi_c = \sum_m \frac{a_m}{\beta_m} (e^{-\beta_m \ell_c} - e^{-\beta_m h_c}) \implies a = \text{NNLS}(A, \Phi_+), \quad A_{cm} = \frac{1}{\beta_m} (e^{-\beta_m \ell_c} - e^{-\beta_m h_c}), \quad (19)$$

solved by projected gradient (no shape priors beyond positivity; the fitted spectra are *balanced* — ~ 0.80 – 0.93 of the branching at $\beta_2 = 18.2 \text{ s}^{-1}$ plus genuine slow tails — the allocation MLE never found in three attempts).

Step 3: dispersion-curve selection of n . The single remaining scalar rescales $a \rightarrow (n^*/n)a$. Candidates are evaluated through the *pipeline’s own* stylized-facts estimator (600s calibrated rollouts, equal-duration matching) and n^* minimizes $\sum_{\Delta} (\log F_{\text{sim}}(\Delta) - \log F_{\text{real}}(\Delta))^2$ over $\Delta \in \{1, 2, 5, 10, 20, 50\}$ s. (A standalone branching-representation simulator — immigrants $\sim \text{Poisson}(\mu_0 T)$, each event spawning $\text{Poisson}(a_m/\beta_m)$ children at $\text{Exp}(\beta_m)$ delays — is used for pre-screening; it agrees with the pipeline estimator at SOL’s criticality and reads $\sim 1.8\times$ hot at BTC/ETH’s, hence the pipeline-based final selection.)

Fitted values (final, per asset). Shared β bank as above; a_m at the selected n^* ; n -contributions a_m/β_m in brackets:

	n^*	$\mu_0 = R(1 - n^*)$	a (nonzero entries)		
BTC	0.9925	0.3042	$a_2=16.920$ [.929], $a_6=0.0004$ [.022]	$a_4=0.0160$ [.027],	$a_5=0.0016$ [.015],
ETH	0.995	0.2138	$a_2=15.717$ [.863], $a_5=0.0046$ [.042], $a_6=0.0005$ [.026]	$a_3=0.0523$ [.016],	$a_4=0.0289$ [.048],
SOL	0.99	0.2226	$a_2=14.540$ [.799], $a_5=0.0060$ [.055], $a_6=0.0015$ [.073]	$a_3=0.0617$ [.019],	$a_4=0.0270$ [.045],

Fit cost: ~ 3 CPU minutes per asset; no gradient training of the ground.

5.5 Mark training (the only learned component)

Adam, lr 2×10^{-3} , weight decay 10^{-6} , gradient-norm clip 0.5, batch 64, **seq/stride 4096** (windows ~ 100 s BTC/ETH, ~ 184 s SOL), **48 epochs** (update parity with the seq-1024/12-epoch baseline), categorical cross-entropy on the single active channel, TBPTT lane batching (stream-order lanes; decoder state — including the ground accumulators — carried detached across windows; lanes reset to learned-init/stationary state at zone boundaries), best model by validation NLL, strict non-finite aborts, TF32 on. During mark training the ground may be MLE-trained or frozen — irrelevant to the final model, since the fitted ground replaces it at transplant (rate-neutrality); the shipped **lgm-kf2** uses exactly this recipe’s checkpoints with the Section 5.4 grounds transplanted.

5.6 Simulation and evaluation protocol

Closed-loop rollouts with carried context ($O(1)$ /step state updates), sampler: inversion (thinning is equally exact via the monotone dominating rate). Protocol-side post-hoc rate calibration (bisection

on κ , validation-rate target, verified within 5%) is retained and lands $\kappa \in [1.01, 1.02]$ — the pin holding in practice. Facts on $600\text{s} \times 32$ -sequence rollouts, equal-duration matched against real segments, 3 rollout seeds per checkpoint.

5.7 What is set vs learned

mean rate	set (pin, algebraic identity)
branching n	measured (regression), then set (dispersion-curve match)
kernel spectrum a	fitted convexly to count covariances (OLS + NNLS)
decay bank β , cell grid, δ	fixed (log-spaced; stated above)
mark distribution $p(k u)$	learned (MLE; the one place neural capacity belongs)

5.8 Final results

All nine evaluation tasks ($3\text{ assets} \times 3\text{ mark seeds}$, 3 rollout seeds each) completed. Fano scales $\{1, 2, 5, 10, 20, 50\}$ s, equal-duration-matched real in the second row of each cell:

	acc (3 seeds)	ppl	time-NLL	κ	Fano model / real
BTC	.428/.428/.429	9.2	−3.425	1.019	[101, 162, 283, 422, 618, 1007] [55, 85, 157, 258, 433, 900]
ETH	.328/.325/.325	14.1	−3.645	1.016	[67, 110, 210, 349, 585, 1170] [79, 125, 245, 417, 748, 1571]
SOL	.194/.196/.195	23.9	−3.035	1.014	[31, 46, 80, 126, 208, 423] [42, 59, 95, 142, 233, 433]

SOL is the clean win: accuracy at or above the SS2P2 baseline (0.192) with the Fano curve within $\sim 25\%$ at 1s converging to $\sim 2\%$ at 50s. ETH matches shape with a modest undershoot (ratio $0.85 \rightarrow 0.74$) and essentially matches volatility clustering (F6 0.512 vs real 0.489). BTC now *over*-disperses at short scales ($\times 1.8$ at 1s, converging to $\times 1.1$ at 50s): the $n^*=0.9925$ log-MSE arbitration ran against a tuning reference hotter at short scales than this eval’s equal-duration real curve; a half-step retune toward $n \approx 0.99$ would center it. The miss is $2\times$ where SS2P2’s was $15\text{--}30\times$ *under*. Two structural signatures confirm the architecture: time-NLL/KS are bit-identical across mark seeds within an asset (the frozen shared ground owns the time law), and short-scale dispersion seed-sd stays at 2–8% (transplant determinism). Excess kurtosis remains the open residual (BTC 47 vs real 88; ETH 7 vs 35; SOL 22 vs 163) — the TL-SSM gate’s target, not the linear ground’s.

6 Outlook: the two-lane state-space point process (TL-SSM)

The LGM-kf architecture runs two exponentially-decaying impulse memories of the same event history (the SSM backbone and the Hawkes accumulators) — sibling objects under different constraints. The unified formulation makes the ground a *reserved lane* of one state machine. State $x(t) = [z(t) \in \mathbb{R}_{\geq 0}^M; h(t) \in \mathbb{R}^H]$ with a shared event loop:

$$\begin{aligned} z(t_i + \Delta) &= e^{-B\Delta} z(t_i^+), & z(t_i^+) &= z(t_i^-) + w_{k_i} \mathbf{1}, & B &= \text{diag}(\beta) \text{ fixed}, & w_k &= \psi(e_k) \geq 0, \\ h(t_i + \Delta) &= e^{-D(h,u)\Delta} h(t_i^+) + \dots, & h(t_i^+) &= h(t_i^-) + B_h e_{k_i} & & (\text{unconstrained}). \end{aligned}$$

Readouts respect the lane disciplines: rate $\Lambda = \mu_0 + a^\top z$ ($a \geq 0$, linear, no normalization — certificate lane); marks $p(k|u) = \text{softmax}(\text{MLP}(u))$ (free lane); and the optional *bounded, mean-one gate* coupling the lanes,

$$\tilde{\Lambda}(t) = \Lambda(t) g(u(t)), \quad g = \exp(\gamma(\tanh(v^\top u) - \bar{c})), \quad \gamma = \log G_{\max}, \quad \bar{c} = \text{EMA}[\tanh(v^\top u)], \quad (20)$$

so g has geometric mean ≈ 1 (pin preserved in expectation; residual absorbed by the verified κ calibration) and $\rho_{\text{eff}} \leq G_{\text{max}}^2 n$ (documented bound). The gate is the *only* path by which the time likelihood reaches the backbone — the fix for LGM’s measured underemployment ($\sim 2\text{pp}$ mark gap) — and its $\text{Var}(g)$ contributes bounded *state-dependent* dispersion (regime burstiness) beyond any linear Hawkes, targeting the kurtosis residual. Gateless ($g \equiv 1$), the TL-SSM is law-identical to LGM-kf; estimation still splits by lane (ground: Kirchner, convex, frozen; h , marks, g : MLE).

Verification before launch (all passed): neutral start $g \equiv 1$ exactly (zero-init v); observed gate range $[0.50, 2.00]$ within the worst-case bound; time-term gradients reach the backbone decays and impulses; frozen ground untouched by optimization ($n = 0.99$ preserved); geometric mean 0.993 after EMA burn-in. Experiment **lgm-tl** (SOL, 3 seeds, seq 4096 / 48 epochs, fitted ground frozen, $G_{\text{max}} = 2$) ran against **lgm-kf2**; the reads were (a) mark accuracy vs the underemployment gap, (b) $\kappa \approx 1$ retention, (c) Fano-curve retention with kurtosis moving toward real.

TL-SSM results (SOL). (a) *Accuracy: pass.* All three seeds reach 0.196–0.197 (vs LGM-kf2 0.194–0.196, ss2p2 0.192), with time-NLL -3.09 to -3.16 (kf2: -3.035) and time-KS 0.19 (kf2: 0.228) — direct evidence the gate restored the time-likelihood gradient to the backbone. (b) *Pin retention: partial fail.* The completed seed needed $\kappa = 1.044$ (kf2: 1.014) and two of three seeds failed the 5% calibration outright (the gated rate is non-monotone in κ near criticality) — the expectation-level pin is measurably looser than the exact pin. (c) *Dispersion: pass.* On the completed seed, Fano $[44, 64, 110, 168, 249, 400]$ vs real $[42, 59, 95, 142, 233, 433]$ — better than kf2 at 1–10s — and excess kurtosis 62 vs kf2’s 22, moving toward the real 163: the $\text{Var}(g)$ mechanism injects the targeted state-dependent burstiness. Net: the gate buys prediction and kurtosis at the cost of pin exactness; calibration robustness (tighter EMA, smaller G_{max} , or a per-seed κ grid) is the outstanding engineering item.

7 The Jain (Compound Hawkes) reference point

Jain’s simulator never optimizes or reports the Fano factor; dispersion is scored via volatility signature plots and $|r|$ -autocorrelation, and his own signature plots undershoot empirical levels by $2\text{--}3\times$ — the published state of the art also misses level while matching shape. What his pipeline gets structurally right:

1. **Estimation by binned-count regression** (Kirchner-style, lin-log lag grid): binning turns the Hawkes process into a vector autoregression on lagged counts whose coefficients are the kernel integrals; the fitting target *is* the cross-scale count autocovariance — the integrand of $F(\Delta)$. Moments, not MLE; the spectrum cannot be misallocated relative to dispersion.
2. Power-law kernels selected by AIC 100% of the time (straight log-log point estimates over six decades of lag, $10^{-4}\text{--}10^2\text{ s}$).
3. Kernel-norm matrix eigenvalue capped at 1 by rescaling (his projection); inhibitory cross-kernels retained.
4. Baselines derived, not fitted: $\mu = (\mathbb{I} - M)\Lambda$ — the matrix form of LGM’s pin.
5. Compound order sizes (Dirac spikes at round numbers + geometric); the constant-size ablation is wrong by $\sim 100\times$ on price variance. Time-of-day multipliers throughout.

The RL chapter’s acceptance checklist. His market-making chapter converts stylized facts into functional requirements, ranked by agent ablations: (1) *clustered arrivals as the agent’s observable* — removing Hawkes intensities from the RL state flips Sharpe $+31.5 \rightarrow -20/-51$, the most catastrophic ablation; (2) *queue dynamics* — removing relative queue position yields pump-and-dump or unprofitable policies; (3) *arbitrage-consistent impact memory* — exponential kernels admit dynamic arbitrage that the optimizer finds (“pump & dump”), rarely so under power-law: mis-specified memory manufactures fake alpha; (4) *responsiveness* — spread widening, queue depletion, liquidity regeneration in reaction to the agent; (5) spread mean-reversion via spread-dependent in-spread intensities ($\propto (s-1)^{0.75}$, $R^2 = 0.85$); (6) realistic frictions (strategies collapse beyond 2 bps). Items (3)–(4) require genuine self-excitation — absent in our measured SS2P2 checkpoints — making the dispersion program a prerequisite for the agent phase, not an aesthetic.

8 Conclusions

1. Prediction and simulation are different objectives; one-step MLE is blind to, and given capacity misallocates, the spectrum controlling $F(\Delta)$.
2. The TBPTT gradient horizon must cover the measured scales: the cheapest large win found (BTC 3/3 seeds transformed).
3. Along one training run, dispersion peaks early and accuracy late; no NLL-selected checkpoint dominates.
4. The rate ceiling is the current level bottleneck (a third of events saturated; teacher-forced dispersion nearly right).
5. Certified dispersion requires linear-in-history structure: only LGM has a settable n , a pinned mean rate, and closed-form $F(\Delta)$ at any horizon; the Kirchner regression is the proven recipe for fitting its ground (typed, per asset) to cross-scale covariances post hoc, marks untouched.
6. Per-asset targets are mandatory (n : 0.948 vs 0.973 is a $4\times$ Fano difference); the tick regime (IS share) should shape typed-kick priors.

Roadmap. (1) cap-9 control (in flight); (2) Kirchner binned-regression fit of the LGM ground; (3) LGM ground $\times$ bounded neural gate ($\rho_{\text{eff}} \leq n G_{\text{max}}$) to restore timing without breaking the certificate; (4) time-of-day multiplier; (5) exploitability audit before trusting any policy trained in the simulator.

Provenance: experiments under `peacock:~/simulation/experiments/ma_cbse/`; code `honglinfu98/simulation` commits `7b97b21..88d22cc`; full lab log in `docs/fano_investigation_2026-08.md` of this repository.